

1. Understanding Asymptotic Notation

Big O Notation

Big O notation describes how an algorithm's running time grows as the input size (n) increases. It helps us compare algorithms and predict their performance for large datasets.

1. $O(1)$ – Constant time
2. $O(\log n)$ – Logarithmic time
3. $O(n)$ – Linear time
4. $O(n^2)$ – Quadratic time

Search Operation Cases

Best Case: The element is found immediately.

Average Case: The element is found somewhere in the middle.

Worst Case: The element is found at the end or does not exist.